

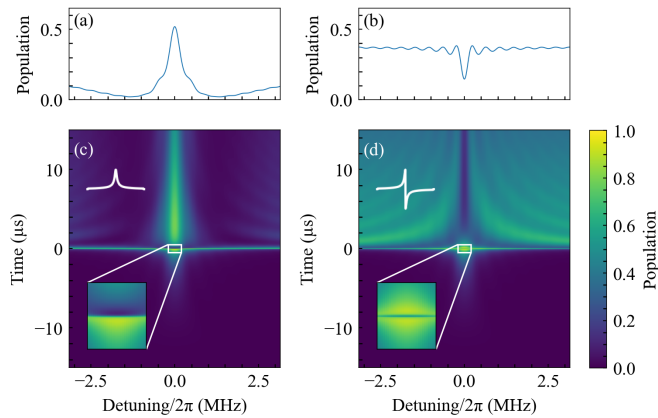


Figure 3. Numerical simulation of qubit spectroscopy under a root-Lorentzian drive and its echo variant. (a) Final excited-state population as a function of detuning  $\Delta/2\pi$  for a standard root-Lorentzian pulse. (b) Same for the echo-root-Lorentzian pulse  $L_{\text{echo}}^{(1/2)}$ . (c),(d) Full time evolution of the excited-state population versus detuning for the corresponding drives. The pulse duration is  $30 \mu\text{s}$  with a cutoff of  $5 \times 10^{-4}$ . While the standard pulse yields a resonant peak at zero detuning, the echo sequence produces an inversion of the central response, leading to a pronounced dip. This behavior reflects cancellation of coherent oscillations at  $\Delta = 0$ , as highlighted in the insets.

adiabaticity breaks down, resulting in a narrowing of the effective spectral response. In contrast, the square pulse is fully governed by conventional power broadening, for which the linewidth increases approximately linearly with the drive amplitude  $\Delta_{1/2} \propto \Omega_0$ . This contrast highlights that the spectroscopic response is not determined solely by the drive strength, but depends sensitively on the temporal envelope of the control pulse, motivating the use of tailored pulse shapes [13–17].

## SPECTROSCOPY RESULTS

Spectroscopy results obtained using the echo-root-Lorentzian pulse, together with standard square and root-Lorentzian pulses for comparison, are shown in Fig. 2. For the measured coherence time  $T_2 \approx 14 \mu\text{s}$ , the corresponding coherence-limited linewidth is  $\Delta_{1/2}/2\pi \approx 20 \text{ kHz}$ . This limit is indicated by a dotted white line in the two-dimensional amplitude-detuning sweep and by gray dashed lines in one-dimensional spectroscopy slices extracted at constant drive amplitude.

Square pulses exhibit pronounced power broadening even at low drive amplitudes compared to Lorentzian-type pulses, with substantial linewidth expansion observed over three orders of magnitude in amplitude. In contrast, the root-Lorentzian pulse displays the characteristic power-narrowing behavior, enabling resolution approaching the  $T_2$ -limited linewidth. However, for pulse

durations comparable to the qubit lifetime  $T_1$ , coherent oscillations arise for both square and root-Lorentzian pulses, resulting in amplitude-dependent linewidth variations that hinder precise spectral extraction.

Introducing a  $\pi$  phase inversion at the peak of the root-Lorentzian envelope partitions the drive into two sequential segments of opposite phase, such that the corresponding forward and backward evolutions cancel exactly at zero detuning. This echo-like structure suppresses resonant Rabi oscillations while preserving the off-resonant spectral response, producing a sharp depletion feature centered at the qubit transition frequency. 1) In the frequency domain, this manifests as a splitting of the spectral response accompanied by a null at resonance, rendering the measured linewidth largely insensitive to drive amplitude. Formally, this mechanism is equivalent to a Loschmidt-echo-type evolution, in which reversibility is achieved only on resonance, while detuning or decoherence breaks the cancellation and restores excitation. Combined with the adiabatic following ensured by the Lorentzian envelope, this interference effect suppresses both power broadening and coherent Rabi oscillations, enabling spectroscopy approaching the  $T_2$ -limited linewidth over a broad range of drive strengths and with pulse durations substantially shorter than those required by standard spectroscopy or Ramsey interferometry.

Fig. 3 shows a numerical simulation of the qubit state evolution during the spectroscopy echo-root-Lorentzian sequence. At resonance, where the adiabatic condition breaks down, coherent Rabi oscillations are induced by the drive. However, due to the  $\pi$  phase inversion at the pulse maximum, the subsequent segment of the drive reverses the accumulated evolution, resulting in a symmetric cancellation that returns the qubit close to the ground state, limited by relaxation ( $T_1 = 30 \mu\text{s}$ ).

For detuned drives, the system remains in the adiabatic regime throughout most of the pulse duration and undergoes excitation only in the vicinity of the pulse peak, where the phase inversion occurs. In this case, the evolution does not cancel and results in a net population transfer followed by relaxation. As a consequence, different detunings produce distinguishable final states, forming the basis of the spectroscopic response.

In Fig. 4, we plot the extracted spectroscopic linewidth obtained from Gaussian fits to the measured response. Although the spectral lineshape of a driven two-level system is often expected to follow a Lorentzian profile, the interference-based structure of the present protocol results in a more complex response. In this regime, Gaussian fitting provides a robust and consistent estimate of the linewidth while yielding values comparable to those obtained from Lorentzian fits within our resolution requirements.

We compare the experimentally acquired linewidth extracted from two-dimensional frequency-amplitude sweeps with numerical simulations of an effective two-